



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Name of Student: Aryan Singh Kasera

PRN: 25070122052

Subject: Programming with Java

Sem: III AY: 2026-27

Assignment No: 12

TITLE :Write a Java program to create and use user-defined packages.

Problem Statement

Write a Java program to create and use user-defined packages.

Learning Objectives

To create and use user-defined packages for organizing Java programs.

Theory

Packages are used to organize related classes and interfaces into namespaces. User-defined packages improve code organization, reduce naming conflicts, and enhance reusability. They also support access protection and modular application development. Java packages simplify project management by grouping logically related components together. Large-scale software systems extensively use packages.

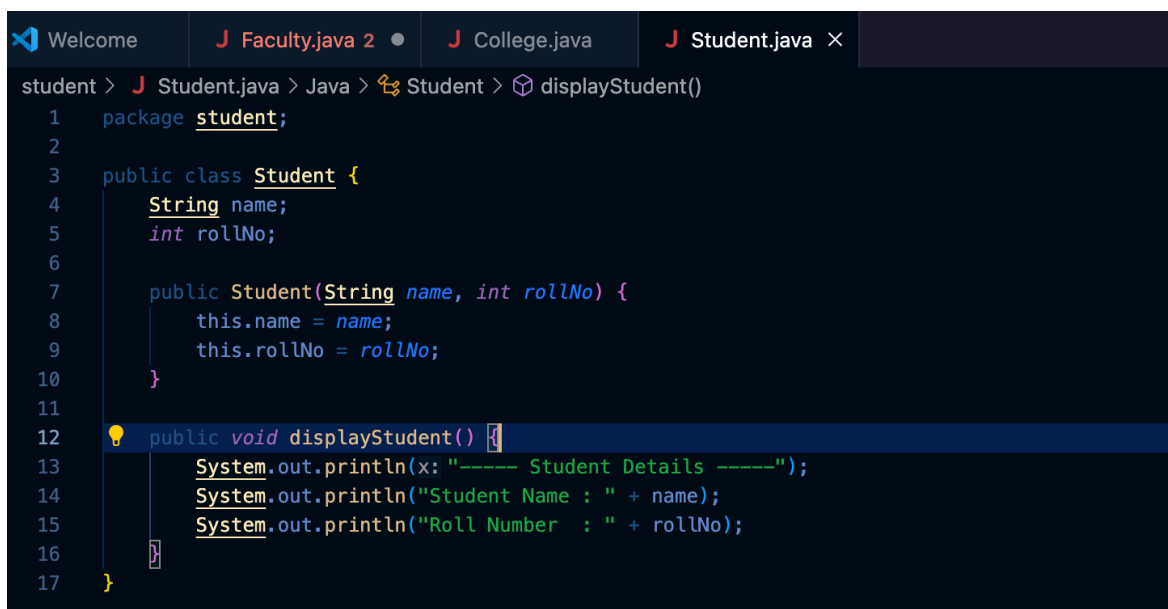
Create a Java program by developing user-defined packages for a College Management System. Create separate packages for student and faculty details, and display their information using classes from these packages.

Faculty Package



```
faculty > J Faculty.java > ...
1  package faculty;
2
3  public class Faculty {
4      String name;
5      String subject;
6
7      public Faculty(String name, String subject) {
8          this.name = name;
9          this.subject = subject;
10     }
11
12     public void displayFaculty() {
13         System.out.println(x: "----- Faculty Details -----");
14         System.out.println("Faculty Name : " + name);
15         System.out.println("Subject      : " + subject);
16     }
17 }
```

Student Package



```
student > J Student.java > Java > Student > displayStudent()
1  package student;
2
3  public class Student {
4      String name;
5      int rollNo;
6
7      public Student(String name, int rollNo) {
8          this.name = name;
9          this.rollNo = rollNo;
10     }
11
12     public void displayStudent() {
13         System.out.println(x: "----- Student Details -----");
14         System.out.println("Student Name : " + name);
15         System.out.println("Roll Number  : " + rollNo);
16     }
17 }
```

College Main file

```
J College.java > Java > College
1  import faculty.Faculty;
2  import student.Student;
3
4  public class College {
5      public static void main(String[] args) {
6
7          Student s = new Student(name: "Aryan Singh Kasera", rollNo: 122052);
8          Faculty f = new Faculty(name: "Dr. Sharma", subject: "Java Programming");
9
10         System.out.println(x: "----- Student Details -----");
11         s.displayStudent();
12
13         System.out.println();
14
15         System.out.println(x: "----- Faculty Details -----");
16         f.displayFaculty();
17     }
18 }
```

Output:

```
----- Student Details -----
----- Student Details -----
Student Name : Aryan Singh Kasera
Roll Number  : 122052

----- Faculty Details -----
----- Faculty Details -----
Faculty Name : Dr. Sharma
Subject      : Java Programming
```

Create a Java program by developing a **user-defined package** named library that contains a class to store and display book details such as book ID, title, author, and price. Import the package into the main program and display the book information.

```
library > J Book.java > Java > Book
1  package library;
2
3  public class Book {
4      int bookId;
5      String title;
6      String author;
7      double price;
8
9      public Book(int bookId, String title, String author, double price) {
10         this.bookId = bookId;
11         this.title = title;
12         this.author = author;
13         this.price = price;
14     }
15
16     public void displayBook() {
17         System.out.println("Book ID : " + bookId);
18         System.out.println("Title   : " + title);
19         System.out.println("Author  : " + author);
20         System.out.println("Price   : ₹" + price);
21     }
22 }
```

```
J Main.java > Java > Main
1  import library.Book;
2
3  public class Main {
4      public static void main(String[] args) {
5
6         Book b1 = new Book(bookId: 101, title: "Java Programming", author: "Arijit Dutta", price: 650);
7
8         System.out.println(x: "----- Book Details -----");
9         b1.displayBook();
10     }
11 }
```

Output

```
----- Book Details -----  
Book ID : 101  
Title   : Java Programming  
Author  : Arijit Dutta  
Price   : ₹650.0
```

Conclusion:

This program demonstrates the creation and use of **user-defined packages** in Java. It shows how packages help organize related classes, improve code reusability, reduce naming conflicts, and make programs more modular, maintainable, and easier to manage.